Investigating the New VLE for the Open University

INTRODUCTION:

In this report, I will examine how the Open University's new Virtual Learning Environment (VLE) affects students' academic performance. The main goals are to see if the VLE helps students improve their grades and whether it's possible to predict grades based on data collected from the platform. To do this, I will analyze data from the Open University Learning Analytics Datasets which contains detailed information about courses, assessments, student demographics, registrations, and interactions with the VLE.

University Background:

The Open University (OU), ranked 11th in the 2021 REF for research impact, is a UK leader in distance education. It offers flexible, innovative programs for diverse students. The VLE leverages technology to enhance learning and align with OU's mission for student success.

VLE Market Overview:

A VLE is an online platform supporting teaching, learning, and assessment. It centralizes course materials, communication tools, and student interaction. Emerging technologies like VR/AR and blockchain are transforming VLEs, emphasizing skill development and accessible learning for all.

Methodology:

I will begin by conducting data wrangling and cleaning processes to prepare the dataset for analysis. This includes handling missing and invalid data, filtering and typecasting variables, and addressing any duplicate entries. Once the data is clean, I will explore trends within the dataset to gain insights into student performance. This exploration will involve identifying the top-performing and least-performing modules, analysing the age distribution of students, and examining weekly activity patterns related to VLE interactions. Understanding these trends will help set the context for evaluating the broader impact of the VLE on academic outcomes.

Further, I will employ statistical hypothesis testing that is **one-tailed, two-sample t-test** to evaluate whether the use of the VLE has a measurable impact on student grades.

The hypotheses tested in this analysis are:

- **Null Hypothesis (H_0):** The Virtual Learning Environment (VLE) does not significantly improve students' grades.
- **Alternative Hypothesis (H_1):** The Virtual Learning Environment (VLE) significantly improves students' grades.

This step will help determine the effectiveness of the VLE in improving academic performance. Additionally, I will build predictive models to forecast student performance based on the available data. Techniques such as feature selection, correlation analysis, principal component analysis (PCA), and linear regression will be utilized to construct these models. Through this process, I aim

to provide actionable recommendations, such as promoting specific VLE features or adapting assessment practices, to help the Open University optimize its VLE and support its mission of fostering academic excellence and inclusivity.

DATA EXPLORATION:

Data exploration is a critical phase in any data analysis process. It involves collecting, cleaning, and analysing datasets to understand their structure, identify patterns, and uncover insights that guide the development of predictive models.

Data Collection and Summary statistics:

For this analysis, seven datasets are utilized:

1. **Courses Data:** Details the structure of Open University modules with module code, presentation code, and duration in days.
   Summary: The dataset includes 22 records across 3 columns with no missing values, covering 7 modules and 4 presentation cycles. Module durations range from 234 to 269 days (average: 256 days), providing insights into timelines.

2. **Assessments Data**: Captures details of assessments, including type (e.g., TMA, CMA, Exam), dates, and weights contributing to final grades.
   Summary: Includes 206 records across 6 columns with no missing values, spanning 7 modules and 4 presentation cycles. Assessment weights average 20.87, ranging from 0 to 100.

3. **Student Assessments Data:** Includes details on individual submissions, scores, and whether assessments were included in final grading.
   Summary: Comprises 173,912 records across 5 columns with no missing values. The average submission date is 116 days, and most scores are not banked (is_banked=1 is low).

4. **Student Data:** Provides demographic and academic details, such as gender, age band, region, previous attempts, education level, and final results.
   Summary: Contains 32,593 records across 12 columns with no missing values. Final results are grouped into four outcomes, with most students passing. The average studied credits are 79.76, and most students have no previous attempts.

5. **Student Registration Data:** Tracks student enrolment and withdrawal dates, highlighting engagement and dropout trends.
   Summary: Includes 32,593 records across 5 columns with no missing values (NaN). The module 'BBB' and presentation cycle '2014J' are most common. Issues include negative date_registration values and many date_unregistration entries marked as '?'.

6. **Student VLE Data:** Records interactions with the VLE, including click counts and interaction dates, to measure engagement.
   Summary: Contains 10,655,280 records across 6 columns with no missing values. The module 'FFF' is most accessed, with '2014J' as the frequent presentation cycle. Clicks

average 3.72 (max: 6,977), and the date field has formatting issues with negative and extreme values.

7. **VLE Data:** Metadata about activities within the VLE, including activity type and time frames.
   Summary: Includes 6,364 records across 6 columns, with no missing values. The most frequent activity type is 'resource', though many week ranges are undefined ('?'), indicating potential gaps in time categorization.

Data Wrangling & Cleaning:

Data cleaning and wrangling are crucial for ensuring datasets are accurate, consistent, and ready for analysis. In this section, I outline the key steps performed and their rationale.

Renaming Columns

I renamed columns to meaningful names for clarity and ease of analysis. For example, 'module_presentation_length' became 'module_length_days', 'date' became 'assessment_date', and 'sum_click' became 'clicks_count', ensuring intuitive, self-explanatory names for seamless analysis and merging.

Handling Invalid Values

I identified instances of the placeholder '?' representing null values in features like 'assessment_score' (173), 'week_from_number' (5243), 'registration_date' (45), 'imd_band_score' (1111), and 'assessment_date' (11). Rows with these values were removed to ensure data integrity.

Filtering Students

For the column 'unregistration_date', the value '?' indicated active students still enrolled in courses, so these rows were retained, while rows with valid unregistration dates (withdrawn students) were excluded. The column was later dropped as it no longer added value to the analysis. Similarly, students with a "Withdrawn" status in the 'final_result_status' column were filtered out to focus exclusively on **active students** who completed their courses. These steps ensured the analysis aligned with the objective of understanding the VLE's impact on engaged students' academic outcomes.

Typecasting Columns

To ensure consistency, ID columns like 'assessment_id', 'student_id', and 'vle_site_id' were typecast from integers to strings, preventing numerical operations and enhancing merges and lookups. Similarly, features like 'assessment_score' and 'week_from_number' were converted from strings to integers for plotting and calculations.

Converting Dates

Columns like 'registration_date', 'submission_date', and 'interaction_date', initially stored as integers relative to a reference point, were converted to calendar dates using '2023-01-01' (e.g., 0 = '2023-01-01', 1 = '2023-01-02', -1 = '2022-12-31') and the pd.to_timedelta function. This transformation enhanced interpretability and accuracy for temporal analysis, improving the clarity of insights.

Removing Duplicate Rows

I identified and removed 787,170 duplicate rows in the student_vle dataset to eliminate redundancy, improve processing efficiency, and ensure the dataset accurately represents unique interactions.

Feature engineering while Merging Datasets

After cleaning, all datasets were merged into a single dataset, consolidating information for seamless exploration and analysis. Feature engineering was applied to enhance predictive power by creating new features:

- **weighted_score** was introduced to standardize assessment scores, calculated as: *[weighted_score = (assessment_score x assessment weights) / 100],* ensuring assessments with varying weights were fairly represented.
- The total clicks per student for each module presentation were aggregated by summing 'clicks_count', capturing overall engagement with the VLE and reflecting learning behaviours.
- **interaction_level** classified students into **'high_interaction'** and **'low_interaction'** groups based on the median 'clicks_count', enabling insights into how engagement levels impact academic outcomes.

The cleaning and wrangling steps resolved inconsistencies, refined column names, ensured accurate dates, and eliminated redundancies, creating a consistent and reliable dataset. The merged dataset is now well-prepared for deeper analysis and robust modelling, providing a solid foundation for insights and predictions.

Exploratory Data Analysis (EDA):

Box Plot for Numeric Data

Box plots visualize numeric data distributions and identify outliers, which can indicate errors or legitimate variations. Understanding the context of each feature is crucial to decide whether to retain or remove outliers. A feature-wise discussion, supported by the box plot (Figure 1), explains the handling of outliers in this dataset.

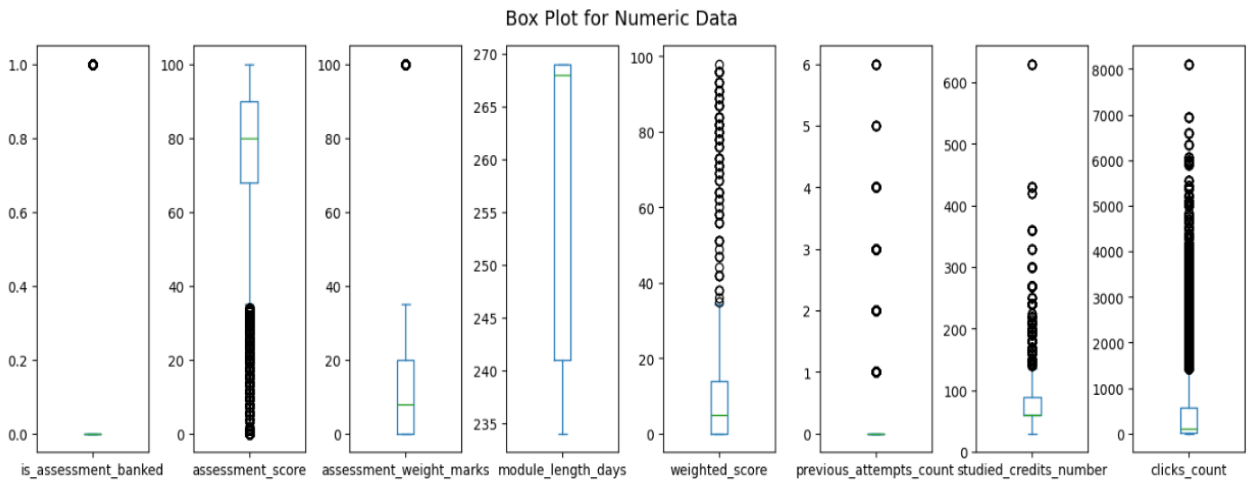

Figure. 1

The column is_assessment_banked contains binary data (0 or 1) and is free from outliers, requiring no additional steps. For assessment_score, a wide range of values, including very low scores, was observed. These low scores may reflect student challenges or circumstances, and removing them could bias the data. Therefore, all scores were retained to preserve the natural distribution (refer to Figure 1).

The assessment_weight_marks column includes valid outliers, such as the value 100, representing full-weight exams. These were preserved as they align with the grading structure (refer to Figure 1). Similarly, the weighted_score column required no modifications as it is a calculated feature.

Outliers in previous_attempts_count likely represent students who struggled or took longer to complete the module, reflecting persistence and resilience. In studied_credits_number, outliers may indicate extensive prior learning or multiple enrolments, offering insights into diverse academic backgrounds. These values were preserved (refer to Figure 1).

Outliers in the clicks_count column reflect variations in engagement styles, such as intensive sessions or frequent short interactions. These were retained as they are essential for understanding engagement patterns (refer to Figure 1 for trends).

By carefully evaluating outliers across these features, meaningful variations in the data were preserved, ensuring the integrity and inclusivity of the analysis.

Relationships Between Demographic Features and Final Result

I used countplots to visualize the relationship between demographic features (student_gender and age_band_years) and final_result_status. Figures 2a and 2b show how final results vary across genders and age bands.

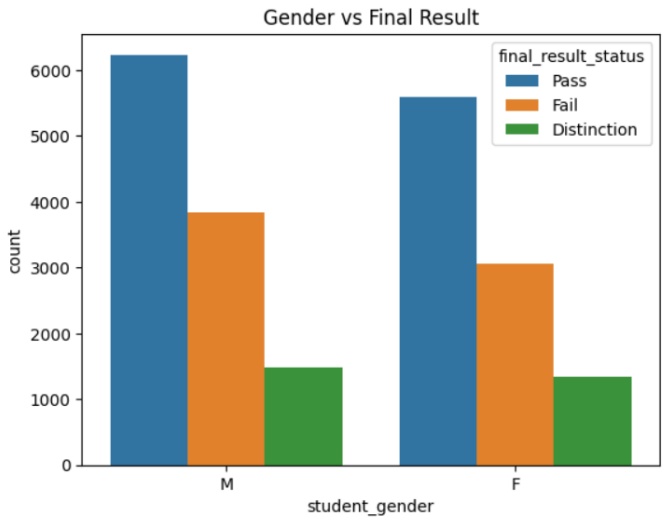

Figure. 2a

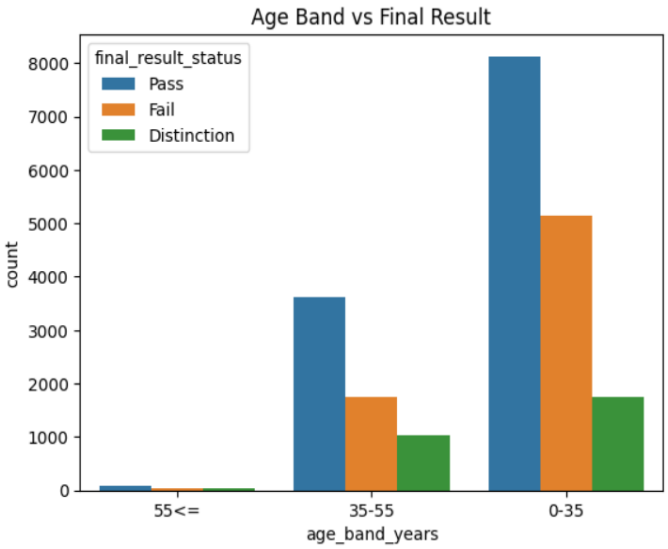

Figure. 2b

Figure 2a shows the distribution of final results (Pass, Fail, Distinction) across male and female students. Although male students are more numerous, a higher proportion of female students fail the course. Distinction rates are similar, but further analysis is needed to assess statistical significance and explore underlying factors.

Figure 2b illustrates variations in academic performance across age groups. The 0-35 group has the highest pass and fail rates, while the 35-55 group shows the highest distinction rate. This suggests age-related factors like prior learning, life commitments, or learning styles may influence outcomes. Further analysis is needed to assess statistical significance and underlying factors.

Top 5 Modules

Understanding the popularity of modules helps identify student preferences and engagement trends within the Open University. Figure 3 presents a bar chart of the top 5 modules based on enrolment numbers, highlighting the most popular modules. It provides a clear representation of module popularity by displaying student counts for each module.

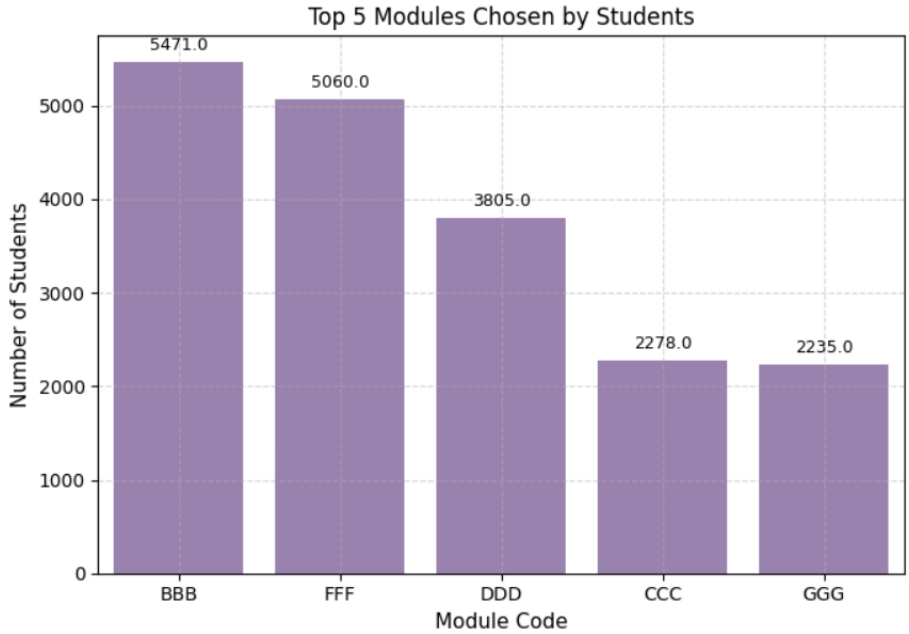

Figure. 3

The bar chart shows the top 5 most popular modules, with BBB leading at 5471 enrolments, followed by FFF (5060), DDD (3805), CCC (2278), and GGG (2235). These insights highlight varying module popularity, guiding resource allocation and support strategies for high-demand courses.

Top 5 Modules with the Highest Average Score

Understanding the modules with the highest average scores provides insights into academic performance across different courses.

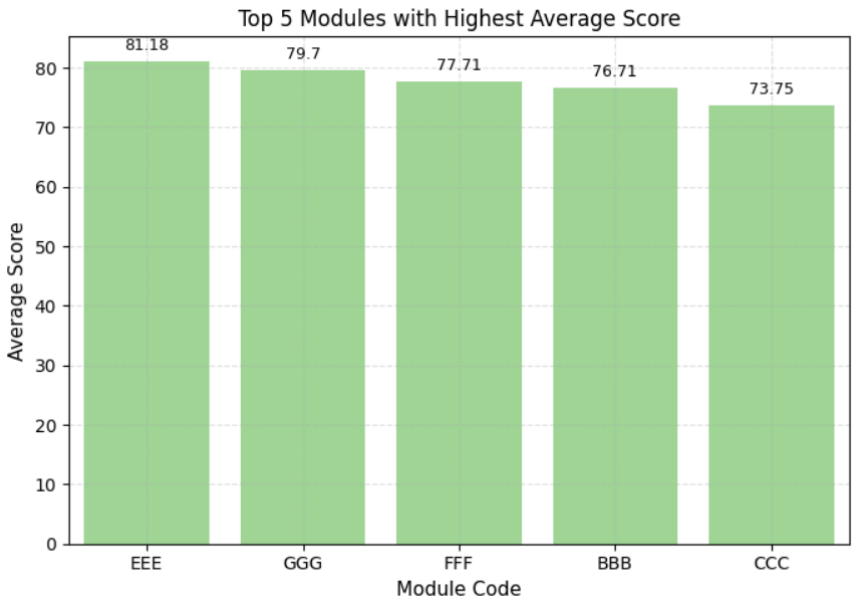

Figure. 4

Figure 4 highlights the top 5 modules with the highest average scores: EEE (81.18), GGG (79.7), FFF (77.71), BBB (76.71), and CCC (73.75). These results suggest better performance in these modules, potentially due to effective teaching, assessment structures, or engagement. This analysis offers insights for enhancing academic strategies and replicating successful practices in other modules.

Top 5 Modules with the Highest Average Score

Analysing modules with the lowest average scores helps identify areas where students face challenges and opportunities for improvement.

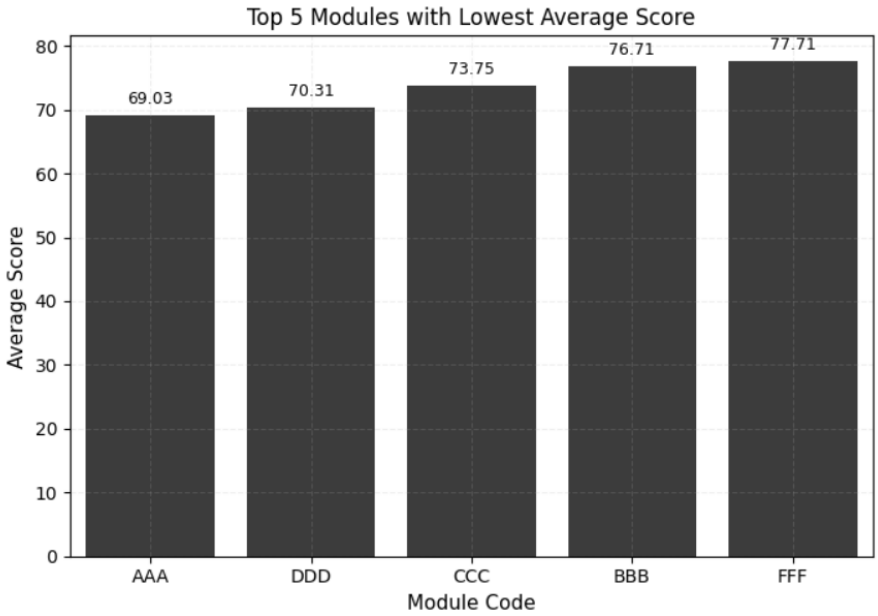

Figure. 5

Figure 5 shows the top 5 modules with the lowest average scores: AAA (69.03), DDD (70.31), CCC (73.75), BBB (76.71), and FFF (77.71). These modules may be more challenging due to content or assessment structure. Identifying such modules aids in designing targeted interventions, like improving instructional methods or revising assessments, to support student success.

Top 5 Modules with the Most Number of Fails

Analysing the modules with the highest number of student failures helps pinpoint courses where additional academic support or curriculum changes may be needed.

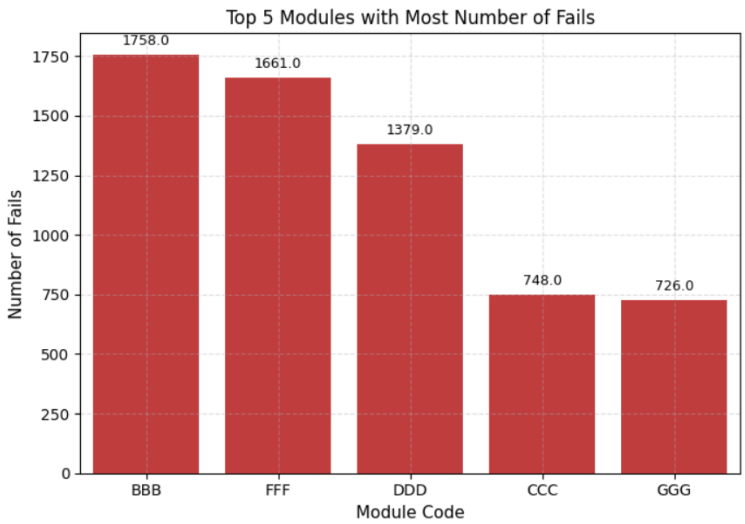

Figure. 6

Figure 6 highlights the top 5 modules with the most student failures: BBB (1758), FFF (1661), DDD (1379), CCC (748), and GGG (726). These challenges may require targeted interventions, such as improved learning resources or alternative assessments, to reduce failure rates and support student success.

Age Distribution of Students

Analysing the age distribution of students provides insights into the diversity of the student population and helps understand the age groups most engaged in learning.

Figure 7 presents the age distribution of students: 69.6% are aged 0-35 years, 29.6% fall within 35-55 years, and 0.7% are 55 years or older. These figures emphasize the significant participation of younger age groups in the Open University's programs.

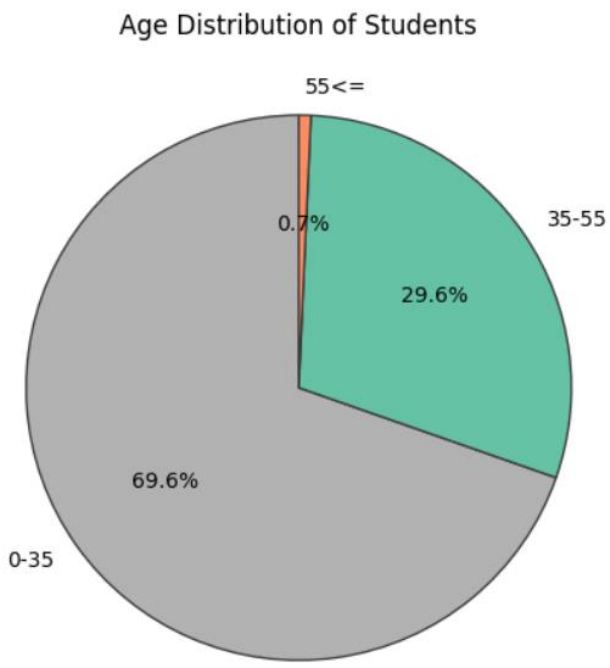

Figure. 7

The data reveals that the Open University caters predominantly to younger learners, with a smaller but notable representation from older age groups, emphasizing the institution's inclusivity and broad appeal.

Weekly Activity Patterns of Students' Interaction with the VLE

Analysing the weekly activity patterns of student interactions with the Virtual Learning Environment (VLE) provides insights into engagement trends over time.

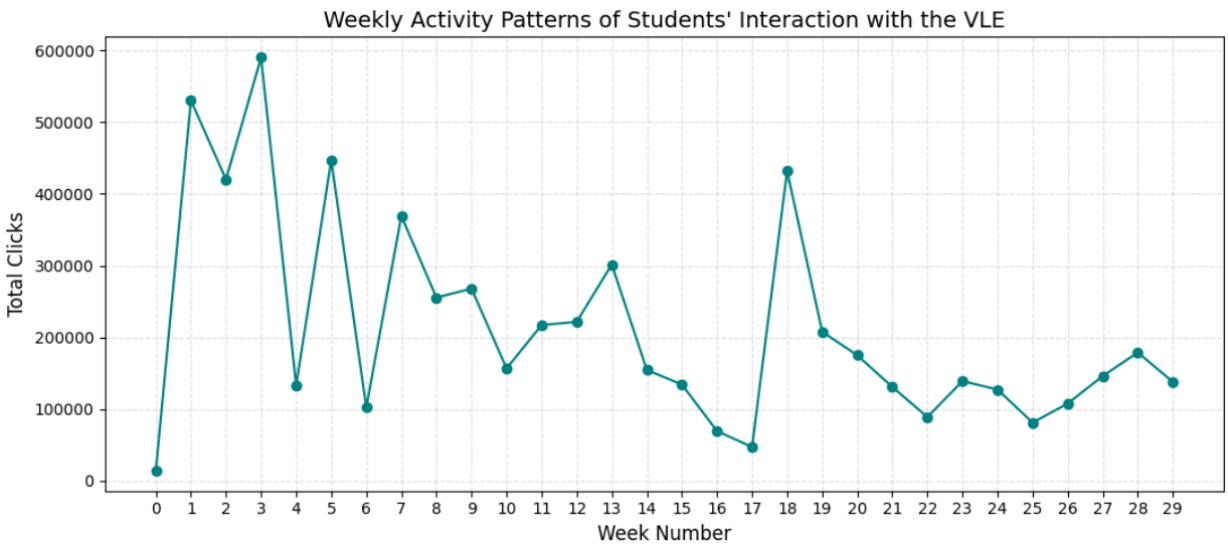

Figure. 8

Figure 8 shows total VLE clicks across 29 weeks. Activity peaks in weeks 2 and 4, likely due to initial engagement and deadlines, with another peak around week 18. Engagement declines in later weeks but remains sustained. These patterns highlight high engagement during early and mid-academic phases, offering insights to optimize VLE resources during critical periods.

METHODS:

Is the VLE improving students' grades?

To evaluate whether the Virtual Learning Environment (VLE) improves students' grades, the feature 'interaction_level' was used as a proxy for student engagement. This classification, based on the median of 'clicks_count', enables the assessment of the relationship between VLE usage and academic performance.

Hypothesis Testing: Impact of Interaction Level on Grades

Hypothesis testing evaluates whether grade differences between high and low interaction groups are statistically significant, **ensuring data-driven conclusions** on the VLE's impact on academic performance and to make decision evidently with 115839 samples.

A **one-tailed, two-sample t-test** was conducted to compare the weighted scores of students with high and low interaction levels. The hypotheses for this analysis were:

- **Null Hypothesis (H_0):** The mean grade of high interaction students is less than or equal to that of low interaction students.
- **Alternative Hypothesis (H_1):** The mean grade of high interaction students is greater than or equal to that of low interaction students.

Hypothesis Testing Results:

The t-test yielded a **T-statistic of 40.65** and a **P-value of 0.00**, which is below the significance level of **5% (alpha = 0.05)**. Consequently, the **null hypothesis was rejected**, indicating that

students with high interaction levels achieve significantly better grades. The confidence interval ranged from **1.97 to infinity**, supporting the conclusion.

The results confirm that students with higher interaction levels in the VLE perform better academically, emphasizing the importance of fostering active engagement with the platform to improve student outcomes.

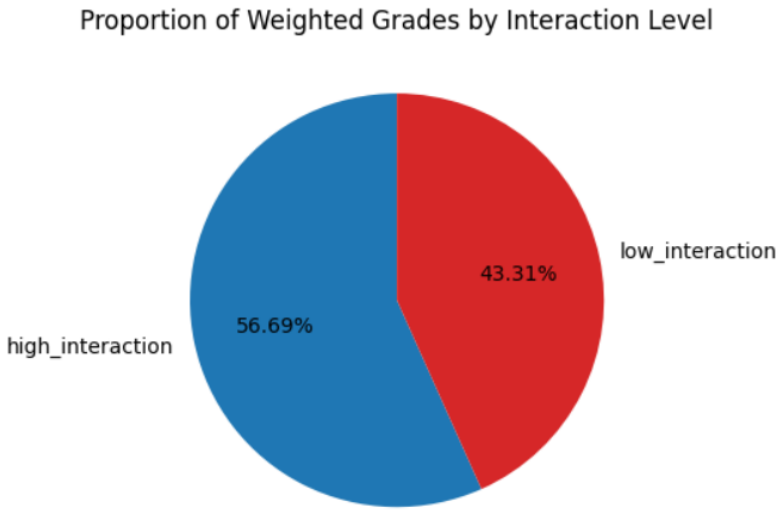

Figure. 9

In figure. 9, the pie chart visualizes the mean weighted grades for students with high and low interaction levels in the VLE. Students with **high interaction** achieved a **significantly higher mean** weighted grade percentage of **56.69%,** compared to **43.31%** for those with **low interaction**. This supports the statistical findings that higher engagement with the VLE positively impacts academic performance, further emphasizing the importance of fostering student interaction to improve outcomes.

<u>Can we predict students' grades?</u>

Feature Selection for Linear Regression Model

To predict students' grades, a linear regression model was developed, involving feature selection, data transformation, and dataset preparation for accurate and robust modelling. I outline the steps taken to prepare the data for analysis.

Dropping Unwanted Columns

To optimize the dataset for linear regression, unnecessary columns were removed. is_assessment_banked, assessment_score, and assessment_weight_marks were dropped as their purpose was captured in 'weighted_score'. Temporal columns like registration_date and unregistration_date were excluded as irrelevant to grade prediction. Unique identifiers such as student_id, assessment_id, submission_date, and assessment_date were removed to avoid unnecessary complexity. Lastly, final_result_status was dropped as it depends on 'weighted_score', making it redundant. These steps streamlined the dataset for modelling.

One-Hot Encoding

Categorical variables were converted to numerical representations using one-hot encoding. Features like module_code, presentation_code, and assessment_type captured course-specific and assessment variations, while student_gender, student_region, highest_education_level, imd_band_score, age_band_years, and has_disability incorporated demographic factors. interaction_level represented engagement. The drop_first=True parameter was applied to prevent multicollinearity, ensuring no redundant information while preserving predictive power.

Pearson Correlation

Pearson correlation measures linear relationships and is commonly used for feature selection in predictive modelling. Unlike Spearman correlation, which measures monotonic relationships, Pearson is better suited for datasets with linear dependencies, like this one. The goal was to identify features significantly correlated with the target variable, 'weighted_score', for inclusion in the regression model.

The features were divided into two sets to improve the clarity of the correlation plots, presented as Figure 10a and Figure 10b, both focused on the target variable 'weighted_score'.

In Figure 10a, strong correlations were observed between:

- The columns 'assessment_type_TMA' and 'weighted_score' (**0.69**) indicating TMA assessments contribute significantly to grades.
- Module-specific features such as 'module_code_EEE' (**0.36**) and 'module_code_GGG' (-**0.31**) also showed a low positive and negative correlation respectively.

However, rest of the features for example 'age_band_years', and 'student_region' showed negligible correlation with the target variable.

In Figure 10b, almost all the features had very weak correlations and negligible correlation with 'weighted_score', confirming they do not significantly influence the target variable.

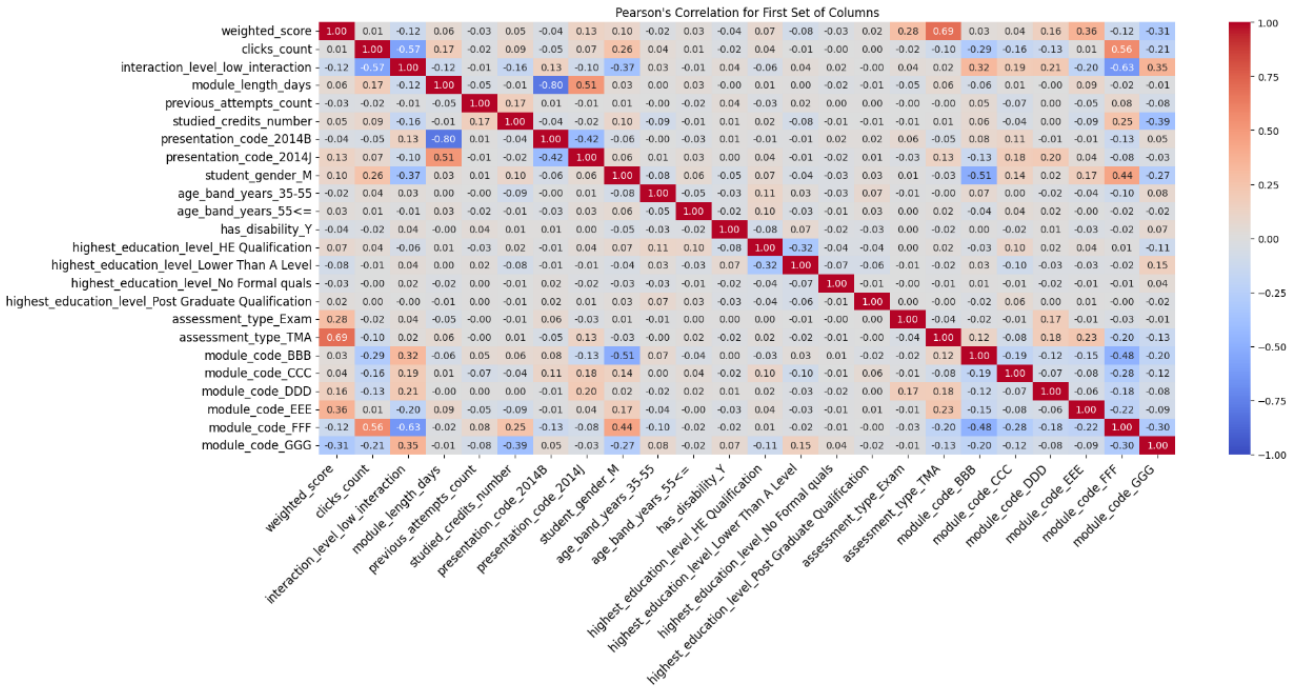

Figure. 10a

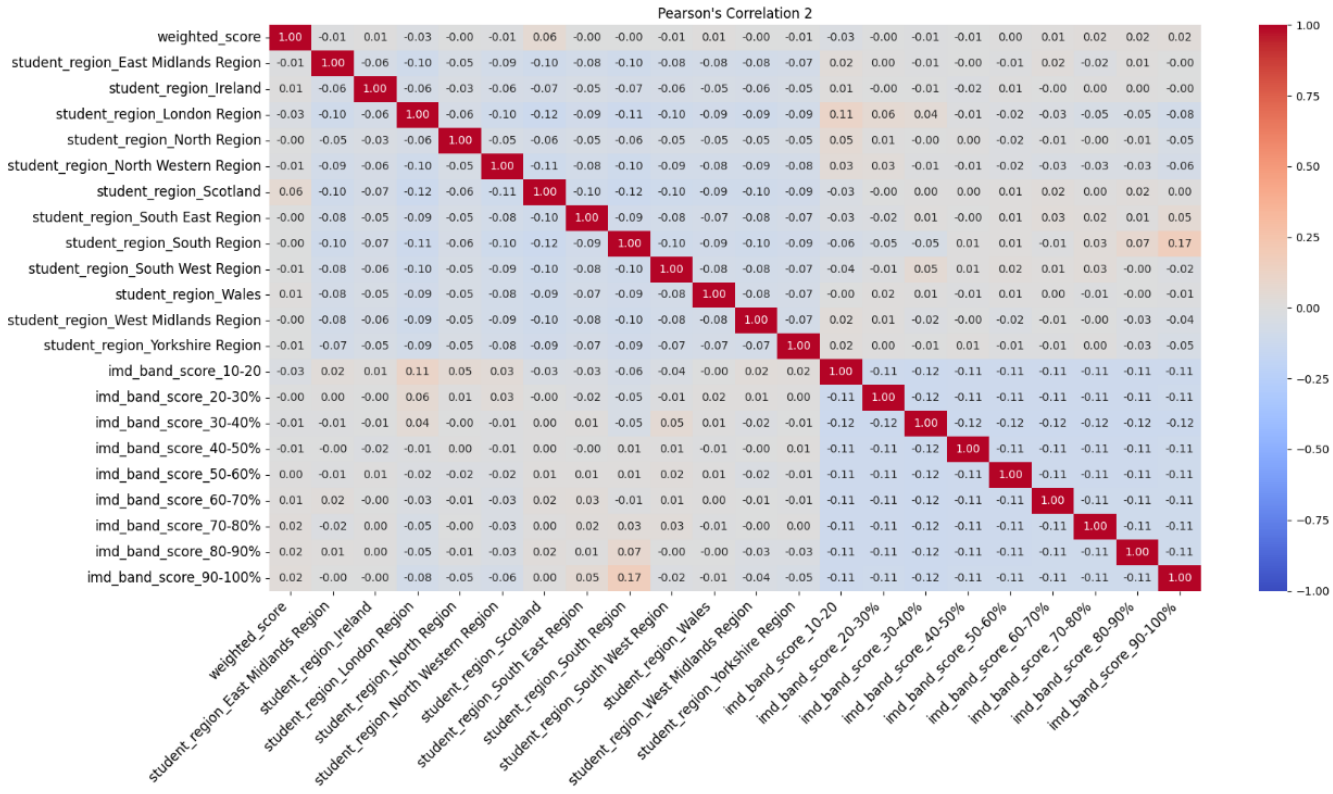

Figure. 10b

OLS Regression

I conducted an OLS regression analysis using the above discussed features that demonstrated at least moderate correlation with the target variable (weighted_score), along with clicks_count. Although clicks_count showed weak correlation, it was included due to its significance in the context of the Virtual Learning Environment (VLE) as a key engagement metric.

The OLS regression model achieved an **R-squared value of 0.57**, explaining 57% of the variance in weighted_score, indicating moderate predictive power. The feature assessment_type_TMA had the strongest positive influence (10.8254), highlighting higher scores on TMAs. module_code_EEE showed a positive impact (7.2976), while module_code_GGG reflected a negative effect (-5.7082) on grades. The feature clicks_count had a negligible positive coefficient (0.0002), suggesting minimal influence on weighted_score despite its relevance in the domain. But still assessment type and module alone cannot be the predictors logically.

Conclusion

The correlation analysis highlighted that very few features had a strong linear relationship with the target variable, 'weighted_score'. Surprisingly, even 'clicks_count', which represents interaction levels with the VLE, showed a weak correlation. This indicates that feature selection based on **Pearson correlation is not feasible** for this dataset. As a result, Principal Component Analysis (PCA) will be applied next to identify key features and reduce dimensionality more effectively.

Linear Regression Model with PCA:

To address the limitations of feature selection using correlation, Principal Component Analysis (PCA) was employed to reduce dimensionality while retaining the majority of the variance in the dataset. PCA transforms the data into a set of orthogonal components, making it ideal for datasets with complex feature interdependencies.

The cumulative explained variance plot (Figure 11) shows how the variance in the dataset is captured by the PCA components. Based on the plot, 95% of the variance is retained with **36 components**, which was determined as the optimal number of components for dimensionality reduction.

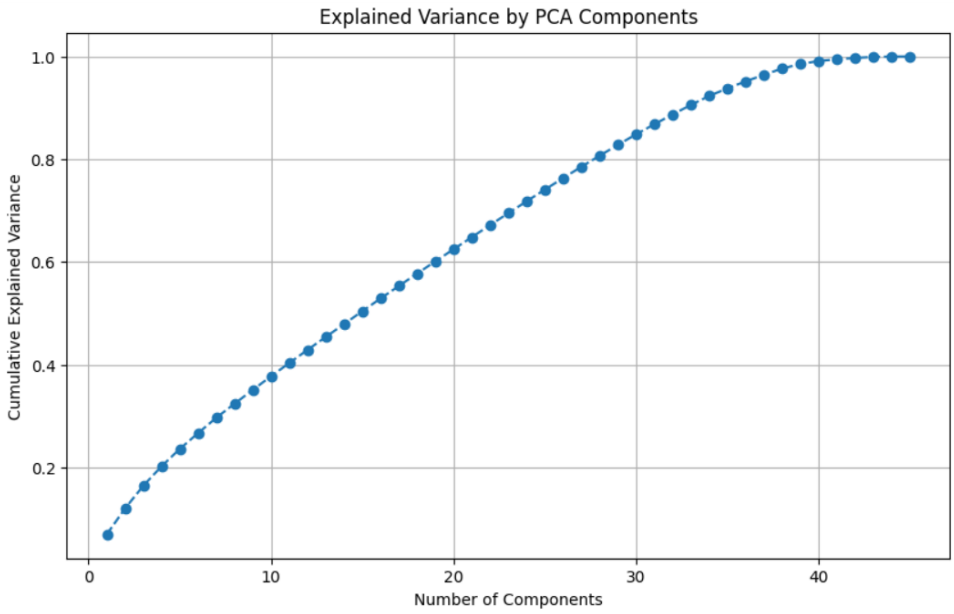

Figure. 11

After applying PCA with 36 components, the dataset was standardized and split into training and testing sets (80:20 ratio) respectively. A linear regression model was trained on the PCA-transformed features. The results highlight the effectiveness of dimensionality reduction in improving the model's efficiency by simplifying the feature set while retaining critical information.

Model Evaluation:

After reducing the dataset dimensions using PCA, a linear regression model was trained and evaluated to predict students' grades. The evaluation metrics and correlation matrix between actual and predicted values provide insights into the model's performance.

Evaluation Metrics:

- **Mean Absolute Error (MAE): 3.76**, indicating the average absolute difference between predicted and actual grades is relatively low.
- **R-squared: 0.67**, showing that the model explains 67% of the variance in weighted_score. This is an improvement compared to the earlier OLS model.

Correlation Matrix:

The correlation matrix (Figure 12) between actual and predicted grades shows a strong positive correlation of 0.82, indicating that the predictions are closely aligned with the actual values.

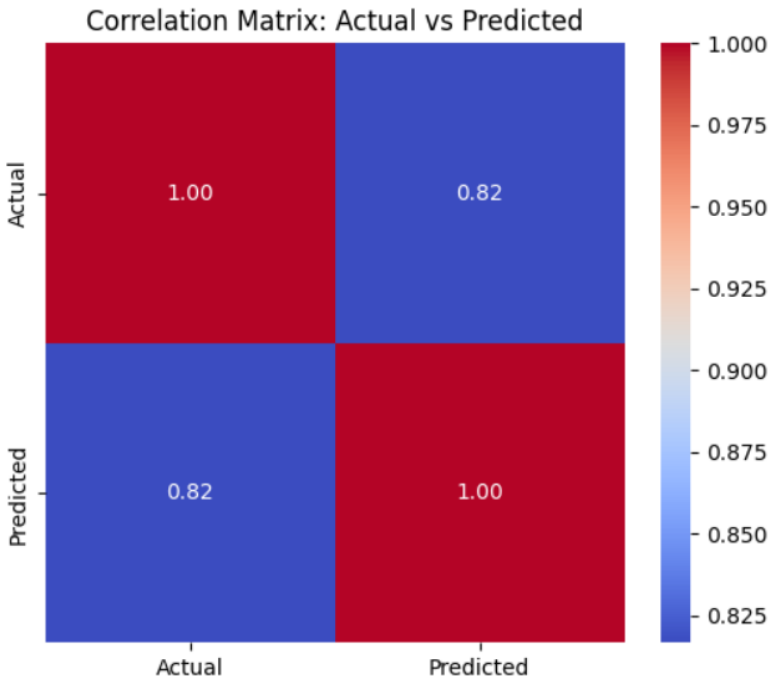

Figure. 12

Results:

The analysis employed various methodologies to evaluate the Virtual Learning Environment's (VLE) impact on students' grades and predict their performance. Key findings are summarized as follows:

- Hypothesis Testing Results: A one-tailed t-test evaluated whether higher VLE interaction improves grades. The null hypothesis (H_0: High interaction does not improve grades) was rejected (T-statistic: 40.65, P-value: 0.00 < 0.05), confirming that higher interaction leads to better grades and highlighting the importance of VLE engagement.
- Feature Selection and Model Calibration: Features were first selected using Pearson correlation, but weak correlations (e.g., clicks_count) led to applying PCA. PCA retained 95% of the variance with 36 components, enabling a robust predictive model.
- OLS Regression Analysis: OLS regression showed an R-squared value of 0.57, explaining 57% of the variance in weighted_score. assessment_type_TMA had a strong positive influence, while module_code_GGG negatively impacted grades. Predictive power was moderate, and clicks_count had a negligible coefficient.
- PCA-Based Linear Regression: OLS regression yielded an R-squared of 0.57, explaining 57% of the variance in weighted_score. assessment_type_TMA positively influenced grades, while module_code_GGG had a negative impact. Predictive power was moderate, with clicks_count showing negligible influence.

Discussion:

The findings confirm that the VLE significantly influences students' academic outcomes, as evidenced by both hypothesis testing and regression modelling. Key insights include:

- **Results in Context:** High interaction levels with the VLE correlate with better grades, as shown by the hypothesis test and PCA-based regression. Additionally, features like assessment_type_TMA were critical predictors, indicating the importance of assessment design in academic success.
- **Limitations:**
    - The PCA approach, while effective for dimensionality reduction, abstracted the original feature relationships, potentially reducing interpretability.
    - Linear regression assumes linear relationships, which may not fully capture complex interactions in the data.
    - Certain features with weak correlations (clicks_count) had limited impact on the models.
- **Actionable Advice:**
    - Actively promote VLE interaction through features like personalized notifications or gamified elements.
    - Investigate and improve modules with high failure rates (e.g., module_code_GGG) and low average scores.
    - Enhance the design of assessments, particularly TMAs, and provide additional support to students struggling in challenging modules.

These recommendations align with the Open University's mission to foster academic excellence through innovative and inclusive learning technologies.

Conclusion:

Summary:

The analysis aimed to evaluate the impact of the Virtual Learning Environment (VLE) on students' grades and predict academic performance. The data was cleaned to address inconsistencies, handle missing values, and prepare it for analysis. Exploratory Data Analysis (EDA) revealed insights into module performance, interaction levels, and student demographics. Interaction levels were used to assess VLE engagement, and hypothesis testing confirmed that higher interaction levels significantly improve grades (T-statistic = 40.65, P-value = 0.00). Feature selection via Pearson correlation highlighted limitations, leading to the adoption of Principal Component Analysis (PCA), which retained 95% of the variance with 29 components. Regression models showed that features like assessment_type_TMA and module_code_EEE positively impacted grades, while clicks_count had minimal influence. The PCA-based regression achieved an R-squared value of **0.67**, with a Mean Absolute Error of **3.76**, demonstrating improved predictive power.

Recommendations:

1. **Enhance VLE Engagement:** Encourage student interaction through personalized notifications, gamified learning elements, and regular feedback mechanisms.
2. **Improve Module Design:** Focus on improving challenging modules like module_code_GGG to reduce failure rates and enhance student outcomes.
3. **Optimize Assessment Strategies:** Leverage insights from assessment_type_TMA to design assessments that better support student learning.
4. **Targeted Support:** Provide additional resources and personalized support to students identified as at risk based on VLE interaction patterns and demographic factors.
5. **Iterative Analysis:** Continuously monitor and evaluate the effectiveness of VLE features and academic interventions using advanced data-driven methods.